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Placental CYP450 gene dysregulation by maternal PFAS: a novel mechanism linking prenatal exposure to altered toddler growth

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Abstract

Background Maternal per- and polyfluoroalkyl substances (PFAS) exposure has been linked to adverse health effects on offspring, but the mechanisms remain unclear. The present study investigates the relationship between maternal PFAS exposure and the expression of placental cytochrome P450 enzymes (CYP19A1, CYP2J2, and CYP2E1), and explores the potential role of these enzymes in linking maternal PFAS exposure to offspring development.

Methods We included 350 mother-infant pairs from the Jiashan birth cohort. Thirteen PFAS compounds were measured in maternal plasma collected at 8–16 weeks of gestation, while the expression levels of the three CYP genes were quantified in placental tissues collected at delivery. Offspring weight and length were measured at birth and at 1, 3, 6, 8, 12, and 24 months of age, and their ponderal index (PI) was computed. Multivariable linear regression was used to examine associations between plasma concentrations of individual PFAS and placental CYP gene expression. Quantile-based g-computation was used to examine the association of the PFAS mixture with placental CYP gene expression. A linear mixed model was used to examine the associations of maternal PFAS and placental CYP expression levels with repeated PI measurements from birth to 24 months. Multilevel mediation analysis was conducted to explore the potential mediating role of CYP genes.

Results Consistent inverse associations between maternal PFAS exposure and the three placental CYP genes expression in female placentas were observed. Specifically, statistically significant decreases were observed in the associations between all PFAS compounds and CYP2J2 expression with β - estimates for the highest exposure versus the lowest from -0.255 to -0.174, as well as between the highest exposure of perfluorotridecanoic acid (PFTrDA) and CYP19A1 expression ($\beta = -0.279$, 95% CI: -0.55, -0.007). In contrast, for male fetuses, significant associations were found between the highest exposure of perfluorohexane sulfonate (PFHxS) and increased CYP19A1 expression ($\beta = 0.404$, 95% CI: 0.131, 0.677), and between the middle of perfluoroundecanoic acid (PFUdA), perfluorododecanoic

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acid (PFDoA), PFTrDA, and perfluoroheptanoic acid (PFHPA) and decreased *CYP2E1* expression ($\beta_{\text{PFUDa}} = -0.286$, 95% CI: -0.511, -0.061; $\beta_{\text{PFDoA}} = -0.270$, 95% CI: -0.495, -0.045; $\beta_{\text{PFTrDA}} = -0.236$, 95% CI: -0.466, -0.006; and $\beta_{\text{PFHPA}} = -0.344$, 95% CI: -0.571, -0.117, respectively). The PFAS mixture was associated with decreased the three CYP genes expression in the placenta of female fetuses, but no association was observed in male placentas. Additionally, placental *CYP19A1* expression was associated with a decrease in girls' PI and an increase in boys' PI, respectively. However, no mediating role of *CYP19A1* in the relationship between PFTrDA and PI in girls or between PFHPA and PI in boys was observed, as the indirect effect was not statistically significant.

Conclusions These results indicate that maternal PFAS exposure may disrupt placental CYP expression and the implications of these disruptions for offspring development warrant further investigation.

Keywords PFAS, Pregnant women, Placenta, CYP expression, Offspring development

Background

Per- and polyfluoroalkyl substances (PFAS) have been widely used for decades in various industrial processes and consumer products because of their unique oleophobic and hydrophobic properties. Since the widespread use of PFAS and their persistence in the environment, virtually all human populations globally, including pregnant women, have been reported to be exposed to some PFAS via different routes, such as oral, inhalation, and dermal routes [1, 2]. Consequently, PFAS have been detected in maternal serum, placenta, cord blood, and fetal organs [3]. Increasing studies have reported that maternal PFAS exposure is associated with various adverse health effects on newborns and their development later in life, such as low birth weight, weakened immune response, and neurodevelopmental problems [4]. Our previous study has found associations between maternal PFAS exposure and increased child ponderal index (in g/cm^3), an indicator of total body adiposity in children [5]. In agreement with our findings, other studies have also found links between maternal PFAS exposure and child adiposity [6]. However, the potential mechanisms by which PFAS exposure links to child development remain unclear.

The placenta plays a critical role in pregnancy by regulating nutrient and oxygen transfer, hormone production, and immune responses [7]. Endocrine disruptors, such as PFAS, may interfere with these functions [6, 8]. Studies on animals have indicated that exposure to PFAS can modify the expression of genes responsible for growth factors, hormone regulation, and cellular signaling within the placenta, which may result in decreased placental and fetal weight [9, 10]. Although recent epidemiological studies, particularly those examining the association between PFAS exposure and epigenetic and transcriptomic changes in the placenta [11, 12], have begun to reveal its potential biological mechanisms, direct evidence linking PFAS exposure to the expression of key functional enzyme systems in placenta, such as the cytochrome P450 (CYP) family, remain incompletely elucidated.

In parallel, CYPs, a family of enzymes involved in the metabolism of both endogenous and xenobiotic substances, play a crucial role in placental function [13, 14]. CYP genes in human placental syncytiotrophoblasts are crucial for placental development and function, as they are involved in numerous metabolic processes [15]. Steroid hormone biosynthesis, for example estrogen synthesis via *CYP19A1*, directly influences fetal adipose tissue development and metabolic programming. Fatty acid and eicosanoid metabolism modulate the inflammatory and metabolic lipid signaling environment within the placenta. Previous studies have indicated that maternal glucocorticoids, drug use, and exposure to environmental pollutants such as phthalates can affect the expression of CYP in the placenta, which may impact normal fetal development [16, 17]. Recent studies found altered gene expression profiles, including CYP genes, in placental trophoblast cells after exposure to perfluorooctane sulfonate (PFOS), perfluorooctanoic acid (PFOA) and a replacement PFAS known as hexafluoropropylene oxide dimer acid (HFPO-DA, tradename GenX) [18, 19]. Other in vitro studies also revealed that PFOS, PFOA and perfluorobutane sulfonic acid (PFBS) inhibit aromatase (*CYP19*) activity [20]. The potential mechanisms remain unclear, but many studies have reported that PFAS are established ligands for nuclear receptors (e.g., *PPAR α* , *CAR/PXR*), which are primary transcriptional regulators of CYP genes. Thus, placental CYP expression represents a sensitive, direct readout of PFAS-activated molecular disruption. However, few studies have investigated the link between maternal PFAS exposure and CYP expression in the human placenta.

In the present study, we examined the association of maternal PFAS exposure during pregnancy with the mRNA levels of three candidate genes in the placenta of term birth, including *CYP19A1*, *CYP2E1* and *CYP2J2*, which were chosen based on a priori knowledge of their expression in the human placenta [21, 22], and their relevance with fetal programming and pregnancy outcomes [15, 23–25]. We further examined the potential role of placental *CYP19A1*, *CYP2E1* and *CYP2J2* in

the association between maternal PFAS exposure and the anthropometry of their children. Importantly, we hypothesized that these associations would differ by fetal sex. This hypothesis is grounded in the established sexual dimorphism of the placenta, where differences in gene expression, metabolic function, and stress responses have been documented. Furthermore, epidemiological studies of prenatal PFAS exposure consistently report sex-specific associations with child growth outcomes [11], such as birth weight and postnatal adiposity, though the direction of effects varies. These observations suggest that the placenta may mediate exposure effects in a sex-dependent manner. Consequently, we designed our analysis to test our primary hypotheses stratified by fetal sex, aiming to identify distinct molecular pathways in male and female placentas that link PFAS exposure to growth.

Methods

Study population

For the present analysis, we used data from mother–infant pairs recruited in the Jiashan Birth Cohort Study, a longitudinal birth cohort conducted in Jiashan County, Zhejiang Province, China. Between September 2016 and April 2018, pregnant women living in Jiashan County were recruited during their first routine antenatal care in the First People's Hospital of Jiashan County and Jiashan Maternal and Child Health Hospital. Detailed information on recruitment and the follow-up profile have been described elsewhere [5]. Briefly, 1,398 pregnant women were recruited at 8–16 weeks of gestation, and 1,040 delivered live singletons at the study hospital. The participants were interviewed in person at recruitment and the scheduled follow-up visits, and children's anthropometric measurements were taken. In the cohort, 942 pregnant women provided sufficient blood samples for PFAS concentrations measurement. Due to limited budgets, mRNA levels of CYP in the placenta were measured in 350 participants. These participants were randomly selected from 893 mother–infant pairs that met the following criteria: 1) maternal plasma PFAS measurement at recruitment; 2) placenta sample at birth; and 3) at least one anthropometric measurement of children. The selection was carried out using the simple random sampling method in SAS software (Fig. 1).

All participants provided informed consent in writing on behalf of their children and themselves. The ethical committee of the Shanghai Institute for Biomedical and Pharmaceutical Technologies approved the study protocol.

Sample collection and preparation

Blood samples were collected at recruitment from the pregnant women during 8–16 gestational weeks. Plasma was isolated from the whole blood within 12 h and was

stored at -80°C with polypropylene centrifuge tubes until shipped on dry ice to the Center for Disease Control and Prevention of Hubei Province for PFAS analysis. Placental tissue samples were collected immediately after delivery. The basal plate (maternal side) and chorionic plate (fetal membranes amnios and chorion) of the placenta were removed. Full-thickness sections were taken from the maternal side of the placenta to the fetal side located 2 cm away from the umbilical cord-insertion site. These sections were washed twice in cold PBS, flash frozen in a dry ice-ethanol bath, transferred to polypropylene containers, and immediately stored at -80°C until analysis.

maternal plasma PFAS measurements

Using high-performance liquid chromatography coupled with tandem mass spectrometry (HPLC–MS–MS; Agilent Technologies Inc., USA), 13 PFAS in maternal plasma including perfluorohexane sulfonate (PFHxS), PFOS, perfluorohexanoic acid (PFHxA), perfluoroheptanoic acid (PFHpA), PFOA, perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), perfluorodecane sulfonate (PFDS), perfluoroundecanoic acid (PFUdA), perfluorododecanoic acid (PFDoA), perfluorotridecanoic acid (PFTrDA), perfluorotetradecanoic acid (PFTeDA), and perfluorohexadecanoic acid (PFHxDA) were measured. The methods of PFAS measurement have been described previously [26].

The limit of detection (LOD) for PFOA, PFDA, PFUdA, and PFDS was 0.01 ng/mL, while the LOD for PFHxS, PFOS, PFNA, PFDoA, PFTrDA, PFHpA, PFHxA, PFTeDA, and PFHxDA was 0.02 ng/mL. Concentrations below the LOD were assigned a value of $\text{LOD}/\sqrt{2}$ in the analyses.

RNA extraction and quantitative real-time polymerase chain reaction (qRT-PCR)

We quantified expression of three genes coding for specific CYP enzymes: *CYP19A1*, *CYP2E1*, and *CYP2J2*. *β -actin* was chosen as housekeeping gene for internal control. The total RNA of 100 mg frozen placental tissues was extracted using TRIzol reagent (Takara, Dalian, China). RNA concentration and quality were assessed using a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific). It is noted that fluorometric assays (e.g., Qubit) can provide more specific quantification of RNA concentration; the use of absorbance-based measurement is a limitation of the current study. Based on the manufacturer's protocol, qualified RNA samples were used to synthesize single-stranded cDNA using a PrimeScript[®] RT reagent Kit with gDNA Eraser (Takara, Dalian, China). RT-PCR amplification was conducted using SYBR Premix Ex Taq[™] (Takara, Dalian, China) on a QuantStudio[™] 6 Flex thermocycler with a 384-well

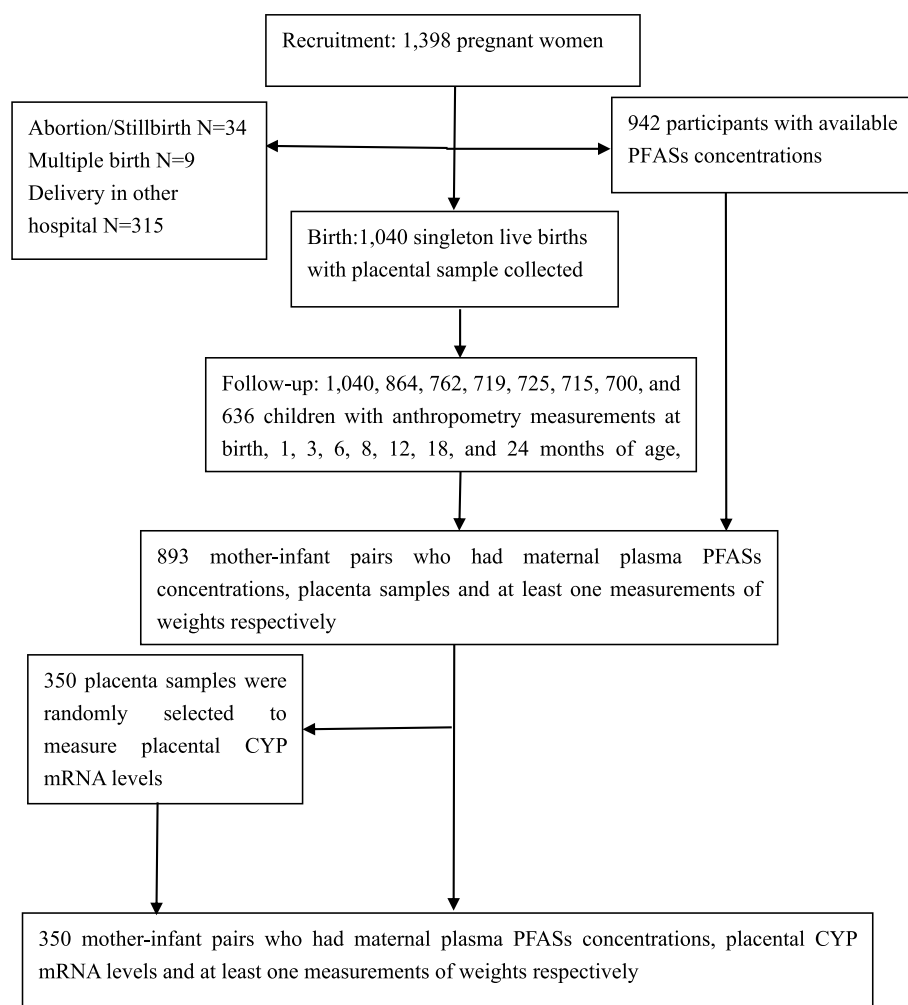


Fig. 1 The study population of the present study

block (Thermo Fisher Scientific) with the following protocol: 1 cycle of initial denaturation at 95 °C for 2 min and 40 cycles of 94 °C for 20 s, 58 °C for 20 s and 72 °C for 20 s. Relative gene expression levels of CYP were calculated using cycle threshold (Ct) values, which was normalized with its housekeeping gene (β -actin) transcript content using $2^{-\Delta C_t}$ method. Three replicates were conducted for each sample, and the average Ct value was calculated. Any assay with no Ct value or a Ct value above 40 was considered “undetectable,” and samples were excluded from that specific gene analysis. Intrarater reliability for the three replicates was high, with intraclass correlation coefficients (ICC) values ranging from 0.97 to 0.99, and it showed excellent consistency for all genes. Proper primer of *CYP19A1*, *CYP2E1*, and *CYP2J2* was designed using Primer 5 (Table S1). The average Ct value, ICC for each analyzed CYP gene across all samples were presented in table S2.

Follow-up and child anthropometry

We collected information on birth weight (g), length (cm), and gestational age from medical birth records. After delivery, children were followed up when they sought routine postnatal health care in a local hospital at ages of 1, 3, 6, 8, 12, 18, and 24 months, where their body weights and heights were measured with a measurement gradation of 0.01 kg for weight and 0.1 cm for length. The Ponderal Index (g/cm^3), calculated as weight (g) divided by height cubed (cm^3), was used to assess early childhood obesity.

Covariates

We administered face-to-face interviews using a structured questionnaire to collect information on maternal age, race/ethnicity, education, income, smoking status, maternal height, pre-pregnancy weight, parity, and other lifestyle factors (e.g., Alcohol Use) at recruitment. Information on pregnancy complications (eg. gestational diabetes mellitus (GDM), Gestational hypertension (GH)),

gestational age, and child sex was collected from the medical records.

The selection of covariates was performed using directed acyclic graphs (DAGs), a straightforward method to show the causal relationships between variables. The minimal sufficient adjustment sets for estimating the effects of PFAS exposure on CYP gene expression were identified based on the web-based software program dagitty.net. (Figures S1), Since the ethnicity of our subjects was exclusively Han Chinese (98.86%) and very few subjects smoked during pregnancy (1.14%), the ethnicity and smoking status were not considered as covariates. Finally, the following factors were included as covariates: maternal age at delivery (<25, 25 to 35, ≥35 years), pre-pregnancy BMI (<18.5 kg/m², 18.5 to 25.0 kg/m², 25 to 29.9 kg/m²), maternal education (middle school and below, high school, college and above), parity (primiparous/parous), gestational age (continuous), and child's sex (male/female) were adjusted for in the analyses.

Statistical analysis

Nine PFAS with detection rates above 90% were included in the analyses. Maternal plasma PFAS concentrations and relative gene expression levels of *CYP19A1*, *-2J2*, and *-2E1* were described using median (inter-quartile range, IQR), and child anthropometry was summarized using mean ± standard deviation (SD). The demographic characteristics and information for both included and excluded mother-infant pairs were summarized using mean ± standard deviation (SD) for continuous variables and frequencies for categorical variables and further compared using student's t test or chi-square tests.

PFAS concentrations and relative gene expression levels of *CYP19A1*, *CYP2E1*, and *CYP2J2* showed skewed (non-normal) distributions and were log₁₀-transformed before subsequent analyses. Pearson correlation was performed to assess the correlations between log₁₀-transformed PFAS concentrations pairs, as well as between the relative gene expression levels of *CYP19A1*, *CYP2E1*, and *CYP2J2*.

We first used generalized additive models (GAM) to explore potential non-linear associations between PFAS concentrations and relative gene expression levels of *CYP19A1*, *CYP2E1*, and *CYP2J2*. An association was considered non-linear if the EDF was > 1.0, supplemented by visual inspection of the smoothed exposure–response curve and some of the associations were found to be non-linear (Table S3). Consequently, we categorized PFAS concentrations into three groups by tertiles to analyze the associations between PFAS and relative gene expression levels through multiple linear regression models, with the lowest PFAS exposure group as the reference. The beta coefficients (β) and 95% confidence intervals (CIs) represent the log₁₀ fold-change in gene expression for each

exposure group compared with the lowest exposure level. Given that previous studies reported the sex-specific effects of PFAS on children's development [27, 28], we introduced a cross-product term for child's sex with each PFAS to evaluate potential interaction effects on placental CYP expression. As several interaction items showed statistical significance ($P\text{-value} < 0.10$), we stratified all analyses by child's sex.

Then, we utilized quantile-based g-computation (QGC) models (R package of qgcomp for quantile g computation) to estimate the joint effect of the PFAS mixture on CYP expression and to assess the contribution of each individual PFAS to the effect. The QGC model provides a g-computation estimator for the mixed exposure via quantile regression, allowing different chemicals within the mixture to be associated with outcomes in different directions [29]. The weight assigned to each chemical indicates its contribution to the joint effect in positive or negative terms. Additionally, the regression coefficients and statistical inferences are derived from 10,000 bootstrap samples, ensuring robust estimates of the effects.

Building on our prior findings demonstrating associations between maternal PFAS exposure and childhood adiposity [5], we further explored the potential mediating role of placental CYP expression in the associations between PFAS and child PI from birth to 24 months. Based on modern approaches of establishing mediation, we employed linear mixed models to additionally examine two longitudinal relationships in mother-infant pairs to identify potential mediators, (1) placental CYP expression levels and repeated PI measurements from birth to 24 months, and (2) maternal plasma PFAS concentrations and repeated PI measurements from birth to 24 months (repeating our prior analyses [5] in this study with a relatively smaller sample size). If statistical significance was found for associations (prenatal PFAS exposure–placental CYP expression; placental CYP expression–child repeated PI) in a chain, it would indicate a potential mediating role of placental CYP between prenatal PFAS exposure and child repeated PI. Linear mixed models allow the inclusion of all available measurements, including pairs with only one or repeated anthropometric measures.

Considering that our longitudinal data can be recognized as a two-level structure, with the repeated PI data during the entire 24-month follow-up period nested within participants, we then conducted multilevel 2–2–1 mediation analysis using Lavaan package of R software [30] to explore the potential mediating role of placental CYP expression in the associations of maternal PFAS exposure and child repeated PI from birth to 24 months. Maternal PFAS exposure and placental CYP expression were modeled at the between-subject level (at level 2) whereas repeated child PI measurements were modeled at the within-subject level (at level 1) (2–2–1 mediation

model) [31–33]. This mediation approach utilizes structural equation modelling and estimates model parameters by means of diagonally weighted least squares with adjusted variance. As is shown in the Figure S2, path (a) and path (b) refer to the indirect effect through placental CYP expression, while path (c') refers to the direct effect of maternal PFAS exposure on child repeated PI. Assessing mediation via path analysis involves defining the direct effect pathway (c') and mediator pathways (a, b) and then uses the product-of-coefficients approach. The analysis provides an estimation of individual regression pathways, along with the indirect effect (ab) and total effect (c' + ab). Bootstrapping was set at 10,000 samples.

To evaluate the impact of extreme outcome values, we conducted a sensitivity analysis by assigning the upper and lower 1% values of the outcome variables to the 99th and 1st percentiles, respectively. Then, we repeated the multiple linear regression to evaluate the associations between PFAS and relative gene expression levels. To directly address concerns about bias from missing values for covariates, missing values for covariates were addressed using multiple imputation with chained equations (MICE) by SAS Proc MI, with the assumption that data were missing at random. A total of 5 imputed datasets were generated, and association analyses were performed based on each imputed dataset separately.

Statistical significance was established at a P -value < 0.05, and analyses were performed using SAS version 9.4 (SAS Institute Inc., Cary, NC, USA) and ggcomp package for quantile g computation, R software version 4.1.3 (<http://www.r-project.org/>).

Results

Characteristics of study participants and placental CYP expression levels

Table 1 displays the demographic characteristics and placental mRNA levels of *CYP19A1*, *CYP2E1*, and *CYP2J2* of the 350 mother-infant pairs included in this study. In the present study, the majority of mothers were aged 25 to 35 years at delivery (70.57%), 61.71% had normal pre-pregnancy BMI (18.5–25.0 kg/m²), and 44.18% had an educational level of college or higher. Almost half were primiparous (49.27%) and only a small percentage smoked (1.14%) during pregnancy. The mean gestational age was 38.96 weeks. The included children were evenly distributed regarding sex (49.14% girls and 50.86% boys). The distributions of most characteristics were generally consistent between the included and excluded mother-infant pairs (Table S4). The expression of *CYP19A1* varies among pre-pregnancy BMI groups, while *CYP2J2* shows variation according to parity. Besides these differences, no statistically significant variations were observed in placental CYP expression levels with respect to maternal age, education, mode of delivery, and child sex.

Maternal plasma PFAS concentration and CYP expression in the placenta

The summarized statistics for PFAS concentration are presented in Table S5. Four PFAS, including PFHxS, PFOS, PFOA, and PFUdA, were detected in all blood samples of pregnant women, with median concentrations of 5.69, 8.57, 11.88, and 3.20 ng/ml, respectively. Five PFAS, PFHpA, PFNA, PFDA, PFDoA, and PFTrDA, were detected in >90% blood samples of participants. The median concentrations ranged from 0.11 ng/mL to 4.58 ng/mL. While PFHxA and PFTeDA were only detected in 10.86% and 31.14% subjects, and PFDS and PFHxDA were not detected in any subjects. There were moderate to strong positive correlations among the nine PFAS congeners concentrations after logarithmic transformation, with Pearson correlation coefficients ranging from 0.31 to 0.88 (P -value < 0.0001; Table S6). *CYP19A1* was detected in all placentas, while *CYP2E1* and *CYP2J2* were detected in 346 out of 350 placentas. The median relative gene expression levels of *CYP19A1*, *CYP2E1*, and *CYP2J2* were 0.1643, 0.0004, and 0.0049, respectively. There were moderate positive correlations between *CYP19A1* and *CYP2J2*, as well as between *CYP2E1* and *CYP2J2*, with Pearson correlation coefficients of 0.48 and 0.46, respectively (P -value < 0.0001; Table S8).

Associations between maternal PFAS exposure and placental CYP expression levels are shown in Table 2. Generally, a consistent pattern of inverse associations between maternal plasma PFAS and CYP expression was observed in female placentas. Especially, all congeners of PFAS were associated with decreased *CYP2J2* expression level, those with of each PFAS was significantly associated with β -estimates for the highest exposure versus the lowest from -0.255 to -0.174 (P -value < 0.05). Additionally, the highest exposure of PFTrDA was significantly associated with decreased *CYP19A1* expression level (β = -0.279, 95% CI: -0.55, -0.007). However, the associations between maternal plasma PFAS and CYP expression in male placentas were less consistent, with statistical significance associations found for the highest PFHxS with increased *CYP19A1* expression level (β = 0.404, 95% CI: 0.131, 0.677), and the middle of PFUdA, PFDoA, PFTrDA, and PFHPA with decreased *CYP2E1* expression level (β_{PFUdA} = -0.286, 95% CI: -0.511, -0.061; β_{PFDoA} = -0.270, 95% CI: -0.495, -0.045; β_{PFTrDA} = -0.236, 95% CI: -0.466, -0.006; and β_{PFHPA} = -0.344, 95% CI: -0.571, -0.117, respectively).

The sensitivity analysis results after removing extreme values were consistent with our primary findings, indicating that extreme values did not significantly affect the observed associations (Table S11). To further examine the stability of our results, missing values for covariates were addressed using multiple imputation. The results using imputed data were generally consistent with our primary

Table 1 Distributions of placental mRNA expression levels of *CYP19A1*, *-2E1*, and *-2J2* among groups with different demographic characteristics in the study ($n = 350$)

Characteristics	N (%) ^a	Placental CYPs mRNA expression[Median (IQR)]					
		<i>CYP19A1</i>	<i>P-value</i>	<i>CYP2E1(E-3)</i>	<i>P-value</i>	<i>CYP2J2(E-3)</i>	<i>P-value</i>
Age at delivery (years)			0.499		0.363		0.493
< 25	83(23.71)	0.204(0.079,0.45796)		0.437 (0.192, 0.968)		4.875 (2.661, 8.829)	
25–35	247(70.57)	0.144(0.051, 0.5709)		0.352 (0.162, 0.672)		4.704 (2.556, 8.857)	
≥ 35	20(5.71)	0.122(0.069,0.885)		0.4214 (0.195, 0.961)		6.992(4.122, 10.923)	
Prepregnancy BMI (kg/m ²)			0.032		0.884		0.341
Underweight (< 18.5)	53(17.43)	0.226(0.102,0.565)		0.376(0.227, 0.834)		4.553(3.055, 10.452)	
Normal (18.5 to 25.0)	217(71.38)	0.123(0.048,0.446)		0.396(0.185, 0.889)		4.768(2.415, 8.859)	
Overweight (25.0 to 29.9)	34(11.18)	0.211 (0.051,0.913)		0.343 (0.192, 1.121)		6.266 (2.686, 11.435)	
Education			0.559		0.901		0.898
Middle school and below	106(31.64)	0.167 (0.049,0.549)		0.407 (0.159, 0.996)		5.064 (2.756, 9.485)	
High school or technical secondary school	81(24.18)	0.165(0.061,0.483)		0.368 (0.162, 0.843)		4.803(3.069, 8.502)	
College and above	148(44.18)	0.160 (0.060,0.628)		0.351 (0.196,0.643)		4.651(2.293, 8.949)	
Ethnicity			-		0.269		0.229
Han	346(98.86)	0.165 (0.056,0.570)		0.359(0.163, 0.728)		4.805 (2.660, 9.123)	
Others	4(1.14)	0.102 (0.073,0.131)		0.957 (0.453, 1.457)		7.341 (3.864, 34.135)	
Parity			0.123		0.418		0.004
Primiparous	173(50.73)	0.1717 (0.063,0.587)		0.376 (0.182, 0.845)		5.755(3.005, 10.708)	
Parous	168(49.27)	0.138 (0.047,0.557)		0.362 (0.162, 0.713)		4.237 (2.496, 7.568)	
Maternal smoking during pregnancy							
Yes	4(1.14)	0.646(0.061,0.615)		0.129 (0.106, 1.873)		3.221 (2.628, 4.155)	
No	202(57.71)	0.1922482(0.075,0.617)		0.363 (0.173, 0.700)		4.893 (2.660, 9.231)	
Gestational week (Mean ± SD, weeks) ^b	38.96 ± 1.07		0.912		0.788		0.672
≥ 37	346(98.86)	0.164 (0.057,0.565)		0.363 (0.163, 0.759)		4.854 (2.660, 9.123)	
< 37	4(1.14)	0.196(0.062,0.696)		0.311 (0.233, 1.443)		4.710 (2.714, 17.870)	
Mode of delivery			0.982		0.358		0.407
Vaginal delivery	232(66.29)	0.159 (0.062,0.455)		0.346(0.175, 0.696)		4.365 (2.660, 8.744)	
Cesarean delivery	118(33.71)	0.180 (0.048,0.668)		0.378 (0.162, 0.879)		5.574 (2.786, 9.654)	
Child sex			0.619		0.576		0.061
Male	172(51.87)	0.156 (0.055,0.605)		0.365(0.175, 0.813)		5.158 (2.881, 9.756)	
Female	178(48.13)	0.167 (0.060,0.549)		3.522 (0.163, 0.708)		4.388 (2.495, 8.716)	

IQR Interquartile range

^aIncomplete data: prepregnancy BMI (46), educational (15), Parity (9), Maternal smoking during pregnancy(144)^bMean ± standard deviation

findings(Table S12), indicating no substantial bias was introduced by the missing data and a high level of confidence in the primary findings.

The results of the QGC analysis are shown in table S9, which were generally consistent with the findings observed in individual exposure-outcome analyses. Statistically significant associations between PFAS mixture and decreased *CYP2E1* ($\beta = -0.109$, 95% CI: $-0.213, -0.005$) and *CYP2J2* ($\beta = -0.163$, 95% CI: $-0.243, -0.083$) levels and a tendency toward decreased *CYP19A1* ($\beta = -0.078$, 95% CI: $-0.229, 0.073$) were found in female placentas. Furthermore, according to the weight of individual PFAS in the associations between mixture and *CYP19A1*, *-2E1* and *-2J2* expression in female placentas (Fig. 2), PFOA (Negative Weight: 0.4331), PFDA

(Negative Weight: 0.2768), and PFDoA (Negative Weight: 0.45118) were the most important drivers for the inverse associations, respectively. While no significant association was found in male placentas, mixed PFAS exposure tended to be associated with increased expression of *CYP19A1*. Additionally, PFHxS exhibited a stronger weight among PFAS mixture, tended to be associated with increased *CYP19A1* expression in male placentas. (Fig. 2).

Associations between maternal PFAS, placental CYP expression and child anthropometry

Summary statistics for child anthropometry at birth, ages 1, 3, 6, 12, 18, and 24 months are presented in Table S10. Table 3 illustrates associations of maternal PFAS exposure

Table 2 Adjusted associations between maternal PFAS concentrations and mRNA expression of CYPs in the placenta at birth (n = 350)

Adjusted β (95% CI) ^{a,b}		Boys			Girls		
		CYP19A1	CYP2E1	CYP2J2	CYP19A1	CYP2E1	CYP2J2
PFHxS							
Lowest tertile	Ref	Ref	Ref	Ref	Ref	Ref	Ref
Middle tertile	0.125(−0.156,0.406)	−0.039(−0.269,0.191)	−0.093(−0.258,0.071)	−0.073(−0.347,0.201)	−0.211(−0.409,−0.013)*	−0.121(−0.271,0.029)	−0.188(−0.343,−0.034)*
Highest tertile	0.404(0.131,0.677)*	−0.096(−0.32,0.128)	0.093(−0.067,0.254)	0.100(−0.182,0.382)	−0.178(−0.382,0.026) [#]		
PFOS							
Lowest tertile	Ref	Ref	Ref	Ref	Ref	Ref	Ref
Middle tertile	−0.151(−0.438,0.136)	−0.119(−0.347,0.109)	−0.07(−0.238,0.097)	−0.185(−0.456,0.085)	−0.16(−0.357,0.038)	−0.146(−0.295,0.003) [#]	−0.174(−0.341,−0.007)*
Highest tertile	−0.032(−0.325,0.261)	−0.204(−0.437,0.028) [#]	−0.047(−0.216,0.122)	−0.206(−0.51,0.098)	−0.145(−0.367,0.077)		
PFOA							
Lowest tertile	Ref	Ref	Ref	Ref	Ref	Ref	Ref
Middle tertile	−0.163(−0.45,0.124)	−0.003(−0.234,0.228)	−0.019(−0.187,0.149)	−0.071(−0.345,0.203)	−0.064(−0.266,0.137)	−0.054(−0.204,0.096)	−0.192(−0.342,−0.041)*
Highest tertile	−0.064(−0.363,0.235)	−0.056(−0.296,0.183)	−0.024(−0.197,0.15)	−0.207(−0.483,0.069)	−0.068(−0.27,0.135)		
PFNA							
Lowest tertile	Ref	Ref	Ref	Ref	Ref	Ref	Ref
Middle tertile	0.083(−0.202,0.368)	−0.168(−0.394,0.058)	0.114(−0.05,0.278)	−0.09(−0.36,0.179)	−0.16(−0.356,0.035)	−0.179(−0.324,−0.033)*	−0.226(−0.378,−0.073)*
Highest tertile	0.038(−0.261,0.337)	−0.086(−0.322,0.15)	0.036(−0.136,0.208)	−0.004(−0.287,0.279)	−0.171(−0.376,0.034)		
PFDA							
Lowest tertile	Ref	Ref	Ref	Ref	Ref	Ref	Ref
Middle tertile	0.087(−0.2,0.374)	−0.16(−0.389,0.069)	0.011(−0.155,0.177)	−0.024(−0.299,0.251)	−0.069(−0.269,0.132)	−0.089(−0.239,0.06)	−0.201(−0.361,−0.04)*
Highest tertile	−0.112(−0.416,0.193)	−0.098(−0.341,0.146)	−0.056(−0.234,0.122)	−0.057(−0.353,0.239)	−0.106(−0.322,0.110)		
PFUdA							
Lowest tertile	Ref	Ref	Ref	Ref	Ref	Ref	Ref
Middle tertile	0.014(−0.273,0.301)	−0.286(−0.511,−0.061)*	−0.042(−0.208,0.124)	−0.083(−0.362,0.195)	−0.171(−0.373,0.03) [#]	−0.108(−0.26,0.044)	−0.192(−0.348,−0.035)*
Highest tertile	−0.015(−0.32,0.291)	−0.138(−0.375,0.100)	−0.041(−0.217,0.135)	−0.116(−0.403,0.171)	−0.207(−0.414,0.001) [#]		
PFDoA							
Lowest tertile	Ref	Ref	Ref	Ref	Ref	Ref	Ref
Middle tertile	0.012(−0.275,0.299)	−0.270(−0.495,−0.045)*	0.007(−0.159,0.172)	−0.194(−0.466,0.079)	−0.024(−0.223,0.175)	−0.103(−0.251,0.044)	−0.255(−0.411,−0.098)*
Highest tertile	−0.008(−0.307,0.29)	−0.091(−0.323,0.142)	−0.015(−0.187,0.158)	−0.128(−0.419,0.162)	−0.144(−0.356,0.068)		
PFTfDA							
Lowest tertile	Ref	Ref	Ref	Ref	Ref	Ref	Ref
Middle tertile	0.001(−0.29,0.293)	−0.236(−0.466,−0.006)*	−0.024(−0.192,0.145)	−0.279(−0.55,−0.007)*	−0.075(−0.275,0.125)	−0.126(−0.277,0.024) [#]	−0.180(−0.338,−0.021)*
Highest tertile	0.086(−0.213,0.386)	−0.123(−0.358,0.112)	0.039(−0.134,0.212)	−0.023(−0.31,0.264)	−0.174(−0.385,0.038)		
PFHpA							
Lowest tertile							

Table 2 (continued)

	Boys		Girls	
	CYP19A1	CYP2E1	CYP19A1	CYP2J2
Middle tertile	0.154(−0.134,0.442)	−0.344(−0.571,−0.117)*	−0.126(−0.393,0.142)	−0.069(−0.215,0.078)
Highest tertile	0.215(−0.068,0.499)	−0.140(−0.361,0.082)	−0.172(−0.445,0.101)	−0.178(−0.328,−0.029)*

*Maternal age at delivery, pre-pregnancy BMI, maternal education, parity, and gestational week were adjusted for in the multiple linear regression

^bEffect estimates (beta coefficients) represent the change in log10—transformed CYP expression for each higher tertile exposure group compared with the lowest exposure level

*Statistically significant differences are marked in bold, *P*-value < 0.05

[#]Marginally significant differences, *p* > 0.05 and *p* < 0.1

and placental CYP expression with child repeated PI from birth to 24 months of age. In girls, we observed that most PFAS, including PFOS, PFOA, PFDA, PFUDA, PFDoA, and PFTrDA, demonstrated a statistically significant association with increased PI, with PFTrDA exhibiting the strongest association ($\beta_{middle \text{ vs. } lowest} = 0.064 \text{ g/cm}^3$, 95%CI:0.029,0.100, $\beta_{highest \text{ vs. } lowest} = 0.069 \text{ g/cm}^3$, 95%CI:0.031,0.106). Similar to girls, most PFAS associated with increased boys' PI. Specifically, higher PFHxS ($\beta_{middle \text{ vs. } lowest} = 0.038$, 95%CI: −0.003,0.079; $\beta_{highest \text{ vs. } lowest} = 0.043$, 95%CI: 0.005,0.080) and PFNA ($\beta_{middle \text{ vs. } lowest} = 0.048$, 95%CI: 0.008,0.087; $\beta_{highest \text{ vs. } lowest} = 0.043$, 95%CI: 0.001,0.086) were linked to increased boys' PI, whereas PFOS was significantly inversely associated with boys' PI ($\beta = -0.048$, 95% CI: −0.088,−0.009), while PFDA and PFUDA showed non-significant inverse associations in the same direction. Additionally, placental *CYP19A1* expression was marginally associated with decreased girl's PI ($\beta = -0.018 \text{ g/cm}^3$, 95%CI: −0.037, 0.001) but increased boy's PI ($\beta = 0.02 \text{ g/cm}^3$, 95%CI: −0.001, 0.042) on a longitudinal scale, while placental *CYP2J2* expression was associated with increased boy's PI ($\beta = 0.061 \text{ g/cm}^3$, 95%CI:0.027, 0.095) (Table 3).

Mediation analysis

The observed associations between maternal PFAS exposure and the placental CYP expression, along with the significant associations of the placental CYP expression with child repeated PI, indicated two potential mediating pathways, PFTrDA-*CYP19A1*-PI in girls and PFHxS-*CYP19A1*-PI in boys. The results of the mediation analyses are shown in Fig. 3. The confidence interval for the indirect effect included zero, indicating no significant mediating effect of *CYP19A1* observed in the associations of PFTrDA or PFHxS exposure with child repeated PI.

Discussion

The present study is the first population—based study to examine the associations between maternal PFAS exposure and altered expression of specific CYP in the placenta. A consistent inverse pattern was observed for maternal PFAS exposure associated with the three placental CYP genes expression in female fetuses, especially with decreased *CYP19A1* and *−2J2* expression. In male fetuses, while significant associations were found between the middle tertile of most PFAS and decreased *CYP2E1* expression, PFHxS was associated with increased *CYP19A1* expression. Additionally, we observed that placental *CYP19A1* was associated with decreased girl's PI, while *CYP19A1* and *CYP2J2* was associated with increased PI in boys. Although no significance reached in the mediation analyses, the

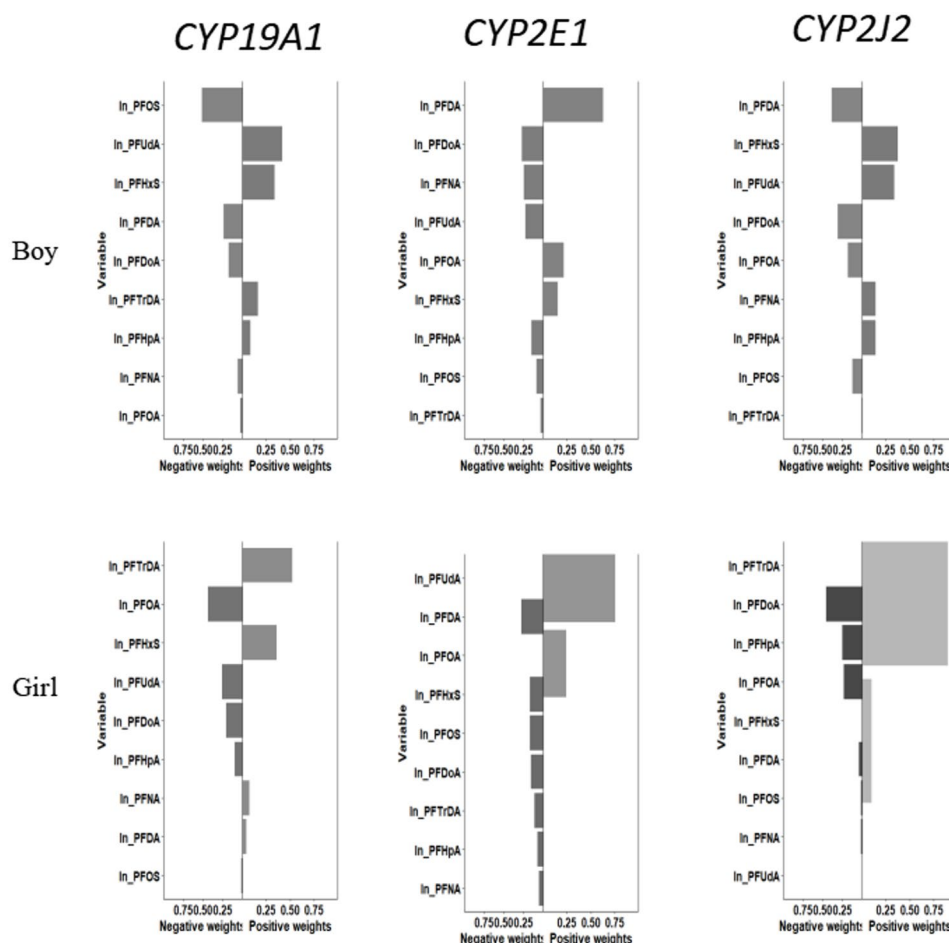


Fig. 2 The directions and magnitude of the assigned weights for each log-transformed PFAS for placental CYP expression in different subgroups. Weights representing relative contribution to the mixture effect of each PFAS component

present study identified potential mediating pathways of PFTrDA-CYP19A1-PI in girls and PFHxS-CYP19A1-PI in boys, which offers an initial insight into the complex underlying mechanisms between PFAS, placental CYP, and children's adiposity.

Previous studies have demonstrated that human cytotrophoblasts cell human cytotrophoblast cell lines—the human choriocarcinoma-derived JEG-3 cells, after exposure to GenX (3 or 300 μ M) or PFOA (100 μ M) for 24 h, exhibit decreased expressions of CYP genes, such as *CYP1A1*, *CYP19A1*, and *CYP2E1* [19, 34], which is consistent with the result of present study. Limited animal studies are also consistent with our study, where they reported that PFOS decreased *cyp17*, *cyp19a*, and *cyp19b* expression after PFAS treatment in zebrafish embryos or mice placenta [9, 35, 36]. Although no previous epidemiological studies addressed the effects of maternal plasma PFAS on placental CYP expression, some studies have shown that the expression of specific CYP in the placenta can change in response to other environmental exposures [16, 17, 37]. For instance, one study has indicated that

maternal exposure to polybrominated diphenyl ethers (PBDEs) was associated with higher expression levels of *CYP2J2* and *CYP2E1* in the placenta [37], which was inconsistent with our results, possibly due to the different physical and chemical properties of PFOS compounds from PBDE and the complex biological mechanisms of PFAS affecting gene expression. Additionally, previous studies have reported PFAS can inhibit CYP activity in vitro [20, 38] and cord serum PFHxS associated with higher placental *CYP19A1* enzymes level in the Laizhou Wan Birth Cohort (LWBC) [39]. Although we did not measure CYP enzymes levels and activities, the present study provided evidence linking PFAS exposure to altered expression profile of *CYP19A1*, *-2E1*, and *-2J2*, which may underlie the changes in CYP enzymes levels and activities.

The mechanisms underlying the associations between PFAS and gene expression in humans are still poorly characterized. However, maternal plasma PFAS exposure is linked to changes in DNA methylation of gene in the placenta [40, 41], potentially affecting transcription

Table 3 Adjusted associations between maternal PFAS concentrations and placental CYP expression levels with repeated measures of ponderal index (PI) from birth to 24 months using linear mixed model [β (95% CI)]

Variables	PI from birth to 24 months of age	
	Boy	Girl
PFHxS		
Lowest tertile	Ref	Ref
Middle tertile	0.038(−0.003,0.079) [#]	0.000(−0.035,0.035)
Highest tertile	0.043(0.005,0.080)*	−0.003(−0.036,0.029)
PFOS		
Lowest tertile	Ref	Ref
Middle tertile	−0.048(−0.088,−0.009)*	0.030(−0.008,0.068)
Highest tertile	−0.028(−0.072,0.015)	0.052(0.011,0.094)*
PFOA		
Lowest tertile	Ref	Ref
Middle tertile	0.007(−0.034,0.047)	0.047(0.012,0.083)*
Highest tertile	0(−0.044,0.044)	0.031(−0.005,0.066) [#]
PFNA		
Lowest tertile	Ref	Ref
Middle tertile	0.048(0.008,0.087)*	0.028(−0.007,0.064)
Highest tertile	0.043(0.001,0.086)*	0.015(−0.022,0.053)
PFDA		
Lowest tertile	Ref	Ref
Middle tertile	−0.034(−0.074,0.007) [#]	0.026(−0.011,0.064)
Highest tertile	−0.03(−0.074,0.014)	0.036(−0.004,0.077) [#]
PFUdA		
Lowest tertile	Ref	Ref
Middle tertile	−0.02(−0.06,0.021)	0.028(−0.01,0.066)
Highest tertile	−0.006(−0.05,0.038)	0.075(0.036,0.115)*
PFDoA		
Lowest tertile	Ref	Ref
Middle tertile	0.026(−0.014,0.066)	0.065(0.029,0.101)*
Highest tertile	0.008(−0.036,0.051)	0.051(0.014,0.089)*
PFTrDA		
Lowest tertile	Ref	Ref
Middle tertile	0.031(−0.011,0.073)	0.064(0.029,0.100)*
Highest tertile	0.037(−0.008,0.081)	0.069(0.031,0.106)*
PFHpA		
Lowest tertile	Ref	Ref
Middle tertile	−0.012(−0.049,0.025)	0.002(−0.032,0.036)
Highest tertile	−0.02(−0.059,0.019)	0.021(−0.013,0.055)
CYPs ^a		
CYP19A1	0.02(−0.001,0.042) [#]	−0.018(−0.037,0.001) [#]
CYP2E1	0.006(−0.019,0.032)	0.006(−0.021,0.032)
CYP2J2	0.061(0.027,0.095)*	−0.01(−0.043,0.022)

Maternal age at delivery, pre-pregnancy BMI, maternal education, parity, gestational age, and child's gender were adjusted for in the model

^alog10-transformed

*Statistically significant differences are marked in bold, P -value < 0.05

[#] Marginally significant differences, $p > 0.05$ and $p < 0.10$

[42, 43]. PFAS may also influence the activity of nuclear receptors like the pregnane X receptor (PXR) and the constitutive androstane receptor (CAR), which are known to play a role in gene transcription [44]. In addition, PFAS may disrupt oxidative stress, and the reactive oxygen species generated from PFAS exposure can regulate gene expression [45, 46]. Further experimental validation studies are needed to verify these mechanisms.

Also, we found that the association between PFAS exposure and placenta CYP expression was sex-specific and non-linear (only observed in tertile 2 in males). Although no previous study has examined the effect of PFAS on CYP in a similar design, studies on individual PFAS compounds did report can disrupt placental growth and function, exhibiting both sex-dependent effects and a nonlinear dose-response characteristic. The underlying mechanisms likely involve PFAS activating or inhibiting nuclear receptors such as PPAR γ and estrogen receptors, leading to downstream transcriptional network alterations that are specific to fetal sex. Additionally, low doses of PFAS may induce sensitive epigenetic reprogramming, while higher doses could trigger cellular stress pathways, resulting in non-monotonic effects.

It is well known that CYP in the placenta, including *CYP19A1*, *CYP2E1*, and *CYP2J2*, play a significant role in the metabolism of steroid hormones and other endogenous compounds such as fatty acids during pregnancy, which are crucial for embryonic development [43, 47, 48]. In general, placental *CYP19A1* converts androgens into estrogens, and the balance of these hormones is a critical determinant of adipose tissue distribution and function [23]. Previous studies have established that reduced CYP19A activity or expression leads to estrogen deficiency and androgen excess—a hormonal profile linked to increased visceral adiposity in both human and animal studies. Therefore, if PFAS exposure downregulates placental *CYP19A*, it could shift the local hormonal balance, potentially programming fetal adiposity in utero [49, 50]. Similarly, placental *CYP2J2* can catalyze the epoxidation of arachidonic acids to produce epoxyeicosatrienoic acids (EETs), disruption of *CYP2J2*-mediated lipid signaling has been associated with obesity and insulin resistance [51]. *CYP2E1* are well-characterized xenobiotic metabolizing enzymes, and are also actively involved in diverse endogenous roles in steroids and other metabolic processes, such as gluconeogenesis [24, 52]. Previous study has also shown that *CYP2E1* is essential to maintain the normal development of human embryos without interference from xenobiotic substances [43]. We therefore propose that PFAS-induced alterations in placental CYP expression may reprogram fetal body composition towards greater adiposity, directly linking our molecular findings to the observed higher Ponderal Index (PI) from birth to 24 months of age.

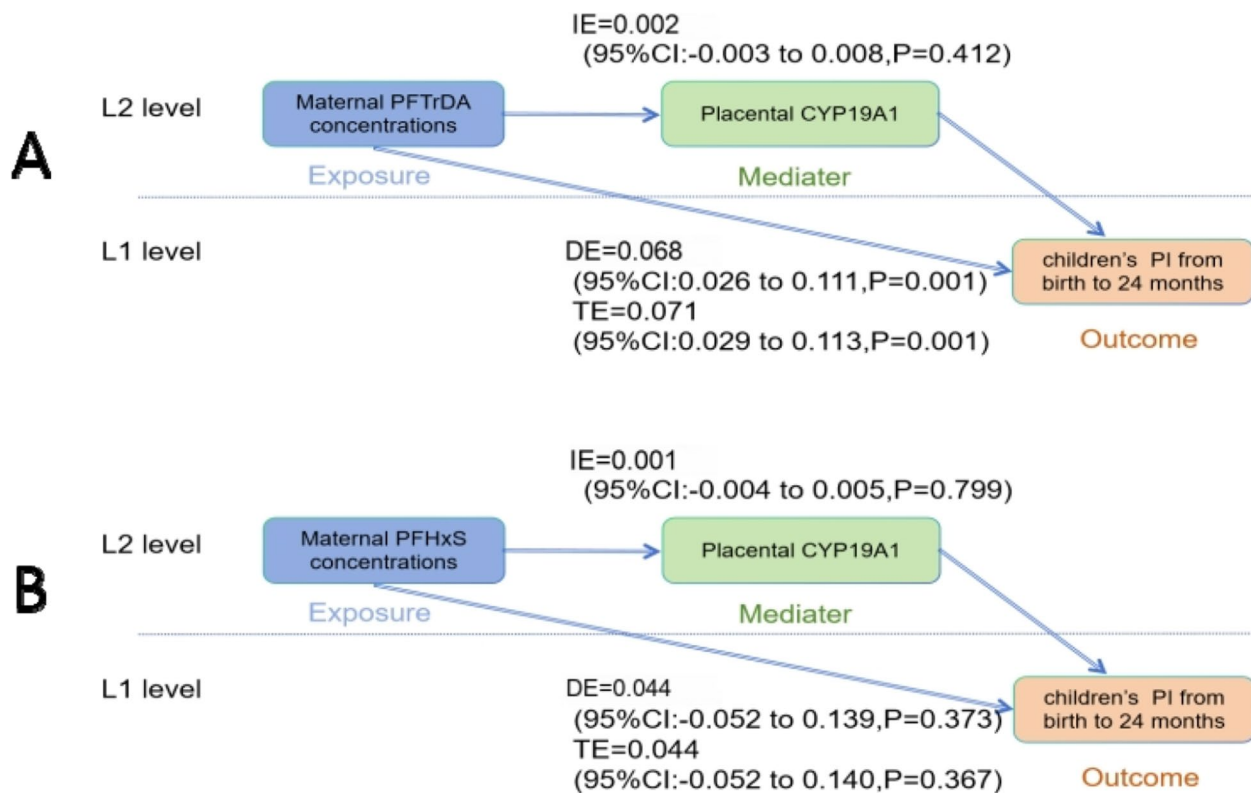


Fig. 3 Mediation analysis for placental *CYP19A1* in the associations between maternal plasma PFAS concentrations and repeated measures of ponderal index (PI) from birth to 24 months. **A** PFTrDA-*CYP19A1*-PI in girls **B** PFHxS-*CYP19A1*-PI in boys. Note: Models were adjusted for maternal age at delivery, pre-pregnancy BMI, maternal educational levels, parity, and gestational age. DE, Direct effect; IE, Indirect effect; TE, Total effect

In the present study, we identified two potential mediating pathways of PFTrDA-*CYP19A1*-PI in girls and PFHxS-*CYP19A1*-PI in boys. However, The mediation analysis didn't reveal statistically significant mediating effects of *CYP19A1*. We should acknowledge that the relatively small sample size in this study may hinder our ability to identify potential mediators. Since *CYP2E1* and *CYP2J2* did not show statistically significant associations across all segments of the hypothesized mediation pathway (i.e., prenatal PFAS exposure-placental CYP expression; placental CYP expression-child repeated PI), their potential mediating effects on the relationship between PFAS and child adiposity were not explored. Considering the observed associations between maternal exposure to PFAS, placental CYP expression, and child's adiposity, as well as the plausibility of mediation, future studies are needed to further explore the mediation hypothesis. Additionally, it is also worth further exploring the potential mediating role of *CYP19A1* in associations between PFAS exposure and other outcomes, such as reproductive development, considering its important role in regulating sex hormones [53].

A key finding of this study was the opposing associations of several PFAS, specifically PFOS and PFDA, with postnatal Ponderal Index (PI)—positive in girls but

inverse in boys. This pattern aligns with epidemiological literature documenting sex-divergent effects of prenatal PFAS exposure on growth and metabolic outcomes, particularly regarding lipid profiles, where associations often differ by sex and obesity status. These opposing associations are biologically plausible, supported by evidence of sex-dependent regulation of placental steroidogenic enzymes (e.g., *CYP19A1*), disruption of lipid metabolism pathways, and differential placental accumulation and transfer of PFAS between male and female fetuses.

Hong et al. (2019) reported that the expression level of *CYP19A1* in term placenta, normalized to β -actin, was approximately 1.5—a value substantially higher than the expression levels we observed for the CYPs examined in our study. This discrepancy is further supported by additional reports in the literature. For instance, studies by Pavék & Smutný (2014) and Berveiller et al. (2022) also describe low or frequently undetectable expression of these CYPs in healthy term placenta, which aligns consistently with our own experimental findings.

Our study has several strengths. First, we utilized a prospective study design that enabled us to examine the long-term association between maternal PFAS exposure, placental CYP, and child adiposity. Additionally, we identified some potential pathways, which provides

preliminary indications of how placental CYP mediate the relationship between maternal PFAS exposure and child adiposity, shedding light on potential mechanisms of PFAS toxicity. Furthermore, our study had more PFAS compounds examined than most prior studies, providing more comprehensive evidence for the health effects of PFAS in the general population.

Several limits of the current study require consideration. First, our sample size was relatively small, which limited our ability to test the placental mediation hypothesis of PFAS exposure and child's adiposity, especially in the context of interactions with sex. Second, we only measured PFAS concentration in maternal blood once. However, owing to the long half-lives of PFAS, its detection in maternal plasma remained stable [54], and a single measurement might have minimal influence on the result. Third, although we adjusted for several important covariates in our study, additional unmeasured or inadequately characterized confounders, such as genetic variations between mother and fetus, and other environmental factors (e.g., GenX and other emerging PFAS), may simultaneously affect PFAS exposure levels and placental gene expression, leading to residual confounding. Fourth, our analysis was only restricted to the subset of women with both placenta samples and anthropometric measurement of children. However, we compared characteristics of participants included in this study with those excluded, and no significant differences were observed. Thus, significant selection biases are not expected. Fifth, we cannot exclude the possibility of the effects of collection conditions on mRNA results, despite our standardized approach for placental tissue collection. And although RNA purity indices were acceptable, the lack of fluorometric quantification (e.g., Qubit) represents a limitation, as NanoDrop may overestimate concentration due to contaminants. Nonetheless, the relative comparisons between exposure groups are unlikely to be systematically biased, as all samples were processed identically.

Conclusions

Our findings suggest that maternal PFAS exposure may disrupt placental CYP expressions, and the latter could be linked with offspring's growth. We also identified two potential pathways, which provides preliminary indications of placental CYP-mediated effects on offspring's growth by maternal PFAS exposure. However, the mediating role of placental CYP expressions warrants further investigation.

Abbreviations

BMI	Body mass index
CYPs	Cytochrome P450 enzymes
LOD	Limit of detection
DE	Direct effect
IE	Indirect effect
PI	Ponderal index

PFAS	Per- and polyfluoroalkyl substances
PFOS	Perfluorooctane sulfonate
PFOA	Perfluorooctanoic acid
PFBS	Perfluorobutane sulfonic acid
PFHxS	Perfluorohexane sulfonate
PFHxA	Perfluorohexanoic acid
PFHpA	Perfluoroheptanoic acid
PFNA	Perfluorononanoic acid
PFDA	Perfluorodecanoic acid
PFDS	Perfluorodecane sulfonate
PFUdA	Perfluoroundecanoic acid
PFDoA	Perfluorododecanoic acid
PFTrDA	Perfluorotridecanoic acid
PFTeDA	Perfluorotetradecanoic acid
PFHxDA	Perfluorohexadecanoic acid
QGC	Quantile-based g-computation
SD	Standard deviation
TE	Total effect

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12940-026-01302-x>.

Supplementary Material 1.

Acknowledgements

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Authors' contributions

Xiuxia Song: Formal analysis, Writing—original draft, Writing—review & editing, Zhaofeng Zhang and Xiaoyu Zhou: Data collection, Methodology, Writing—review & editing, Hong Liang: Funding acquisition, Supervision, Methodology, Writing—review & editing, Yao Chen: Formal analysis, Methodology, Writing—review & editing, Weiqiang Zhu: Methodology, Writing—review & editing, Wei Yuan: Supervision, Methodology, Writing—review & editing, Jing Du: Funding acquisition, Supervision, Methodology, Writing—review & editing, Ziliang Wang: Funding acquisition, Conceptualization, Writing—review & editing, Maohua Miao: Supervision, Resources, Writing—review & editing.

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Data availability

The data that support the findings of this study are available upon request and with the permission of the authors.

Declarations

Ethics approval and consent to participate

The study protocol was approved by the ethical committee of the Shanghai Institute for Biomedical and Pharmaceutical Technologies, and the research was conducted in accordance with the Declaration of Helsinki. All participants provided informed consent in writing on behalf of their children and themselves.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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